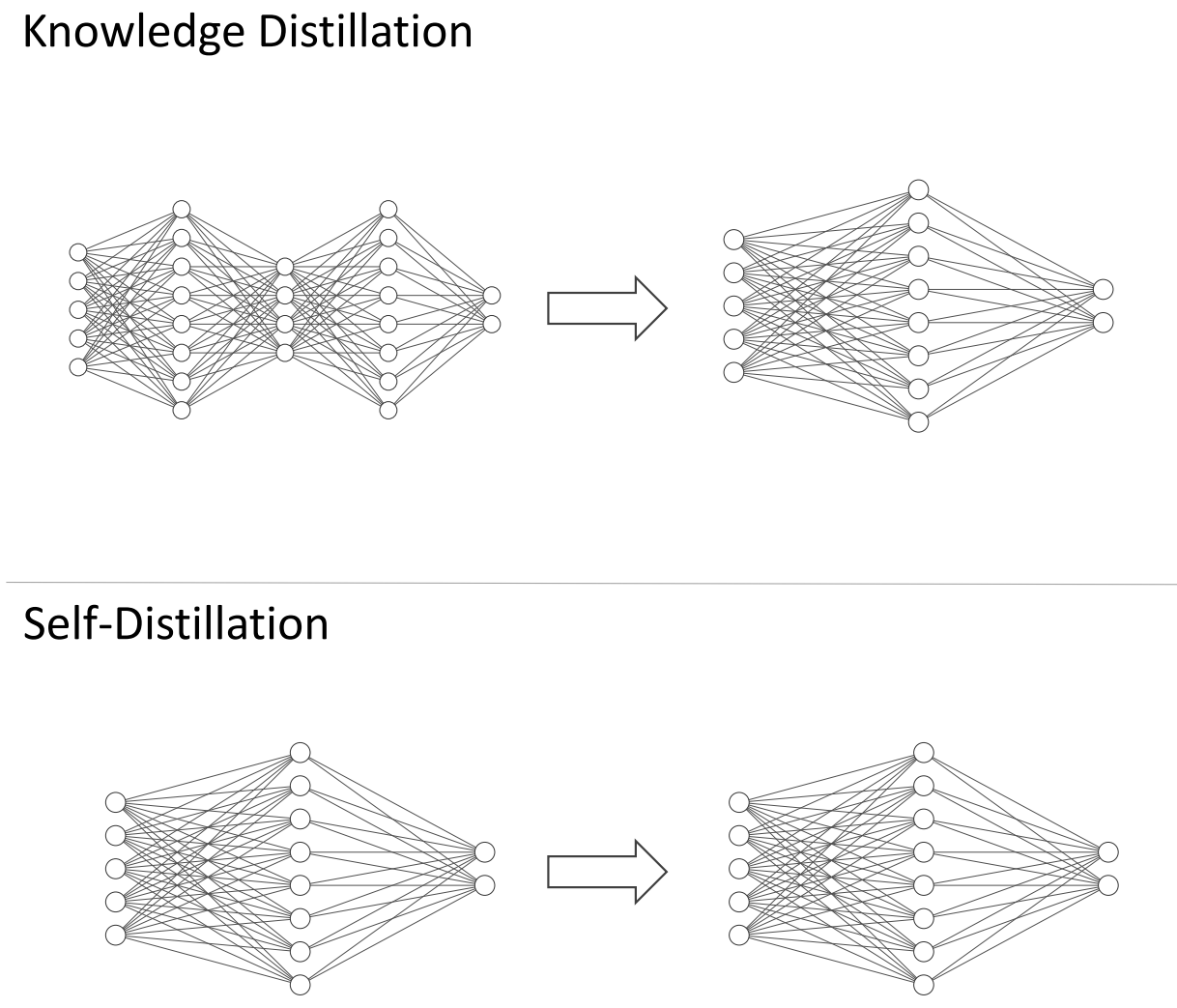


Knowledge Distillation and Self-Distillation

Knowledge Distillation (KD)

- **Teacher model:** Large, costly model. (e.g., ensemble)
- **Student model:** Small, cheap model.
- KD loss = $(1 - \xi) \cdot \mathcal{L}_{\text{CLS}} + \xi \cdot \mathcal{L}_{\text{Distill}}$
 - $\mathcal{L}_{\text{CLS}}$: Loss btw. student prediction and target
 - $\mathcal{L}_{\text{Distill}}$: Loss btw. student prediction and teacher prediction
 - ξ : Loss weight
- **Transferring Dark Knowledge**
 - Class correlation, Sample weight



Self-Distillation (SD): Teacher and student share the same architecture.

Why does SD work? What is the “dark knowledge” transferred from teacher to student?

Setup and Preliminaries

Task: K -class classification with fixed feature extractor (i.e. linear probing)

- Dataset: Kn feature-label pairs, $\{\phi(\mathbf{x}_i), \hat{y}_i\}_{i=1}^{Kn}$. $\hat{y}_i$: given label, y_i : ground-truth label
- Classifier: $\mathbf{y} = \sigma(\boldsymbol{\theta}^\top \phi(\mathbf{x}))$, $\sigma: \mathbb{R}^K \rightarrow (0, 1)^K$: softmax function.
- **Multi-round self-distillation:** t -th model targets the outputs of $(t-1)$ -th model.
- Training Objective: ℓ_2 -regularized cross-entropy (CE) loss

$$f^{(t)}(\boldsymbol{\theta}) = \frac{1}{Kn} \sum_{i=1}^{Kn} \underbrace{\text{CE}(\mathbf{y}_i^{(t-1)}, \mathbf{y}_i^{(t)})}_{\text{targets } (t-1)\text{-th model as targets}} + \frac{\lambda \|\boldsymbol{\theta}\|_F^2}{2},$$

where $\mathbf{y}_i^{(0)} := \mathbf{e}(\hat{y}_i)$.

- Softmax output matrix of t -th distilled model: $\mathbf{Y}^{(t)} = [\mathbf{y}_1^{(t)}, \dots, \mathbf{y}_{Kn}^{(t)}] \in \mathbb{R}^{Kn \times Kn}$
- Gram matrix $\Phi \in \mathbb{R}^{Kn \times Kn}$: $[\Phi]_{ij} := \langle \phi(\mathbf{x}_i), \phi(\mathbf{x}_j) \rangle$

Label Averaging Effect of Self-Distillation

Theorem 3.1 (Label averaging effect of SD)

As the number of self-distillation rounds $t \in \mathbb{Z}^+$ increases, the output $\mathbf{Y}^{(t)}$ of the t -th distilled model evolves as

$$\mathbf{Y}^{(t)} - \frac{1}{K} \mathbf{1}_{K \times Kn} = \underbrace{\left(\mathbf{Y}^{(0)} - \frac{1}{K} \mathbf{1}_{K \times Kn} \right)}_{\text{Given label matrix}} \underbrace{\left[\mathbf{I}_{Kn} - \left(\mathbf{I}_{Kn} + \Phi / (K^2 n \lambda) \right)^{-1} \right]^t}_{\text{Label averaging matrix}}$$

Intuition: Label averaging matrix $\Phi^{(t)} := \left[\mathbf{I}_{Kn} - \left(\mathbf{I}_{Kn} + \Phi / (K^2 n \lambda) \right)^{-1} \right] \approx \Phi / (K^2 n \lambda)$

→ Self-distillation can be interpreted as **label averaging** based on the Gram matrix.

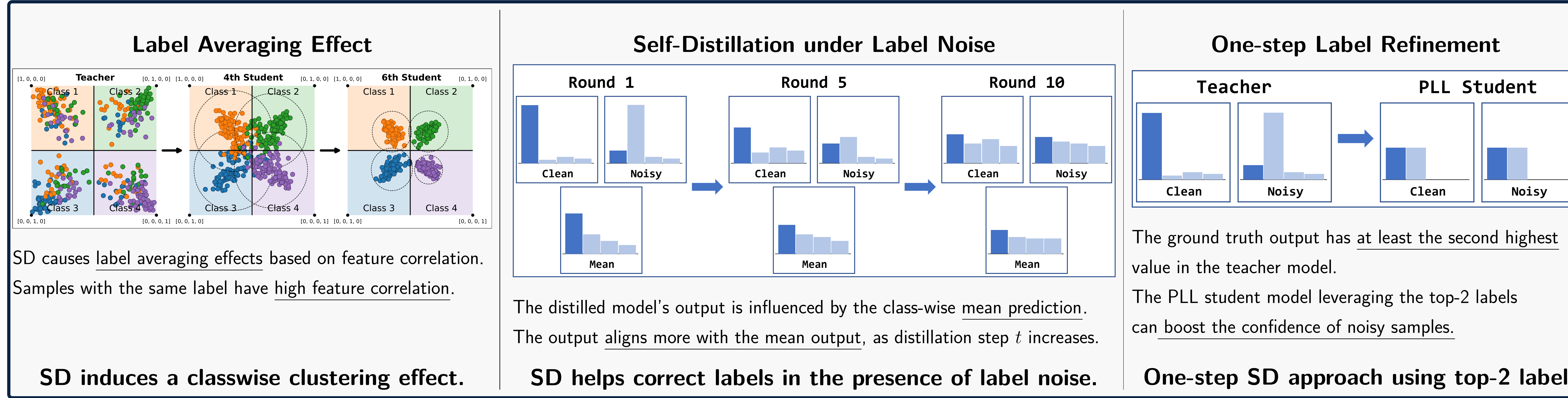
Eigendecomposition of label averaging matrix $\Phi^{(t)}$

$$\Phi = \sum_{i=1}^{Kn} \lambda_i \mathbf{v}_i \mathbf{v}_i^\top, \quad \Phi^{(t)} = \sum_{i=1}^{Kn} \left(\frac{\lambda_i}{K^2 n \lambda + \lambda_i} \right)^t \mathbf{v}_i \mathbf{v}_i^\top$$

- $\Phi^{(t)}$ shares the same eigenvectors with the Gram matrix Φ .
- The eigenvalues of $\Phi^{(t)}$ are ordered in the same way as the eigenvalues of Φ .
- As t increases, label averaging effects of SD are dominated by the top eigenvectors.

How many eigenvalues become dominant?

Overview

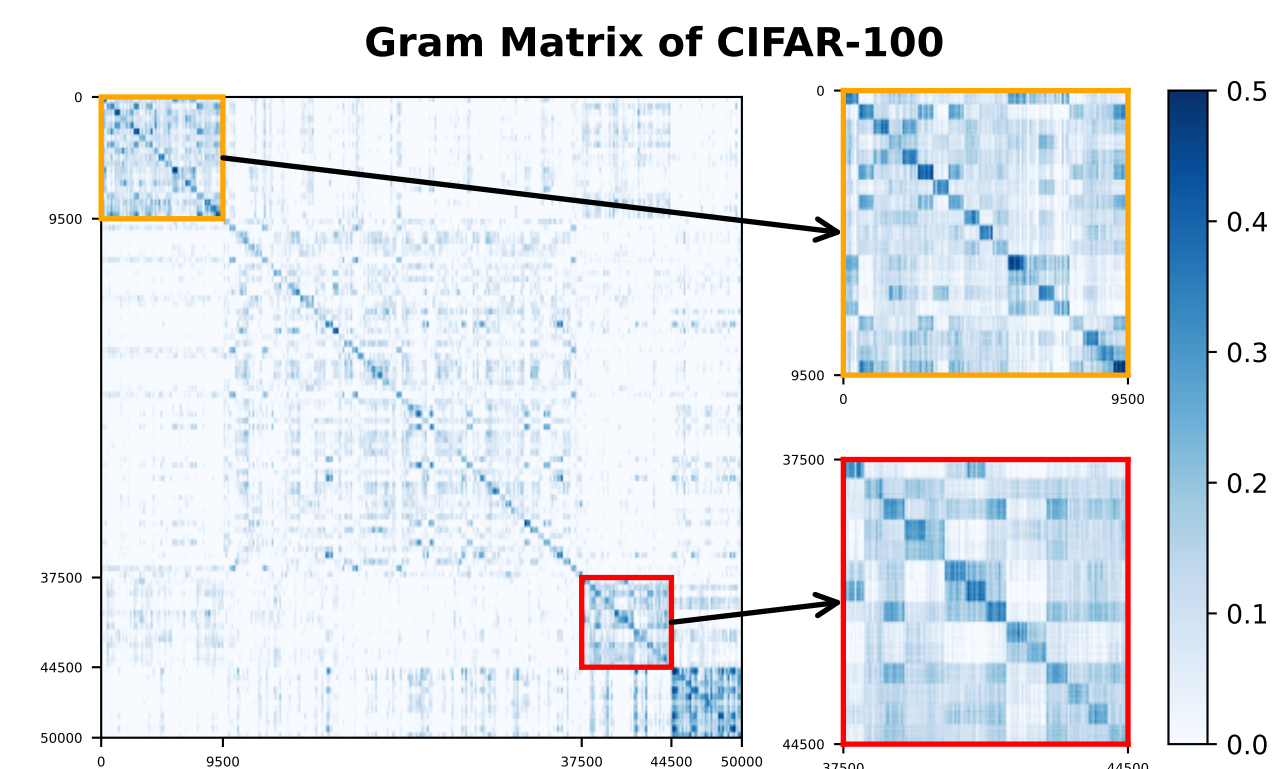


Low-Rank Structure for the Gram Matrix

$$\Phi = \underbrace{\begin{pmatrix} c & & & \\ & d & & \\ & & d & \\ & & & d \end{pmatrix}}_{\text{class correlation matrix}} + \underbrace{\begin{pmatrix} & & & \\ & & & \\ & & & \\ & & & \end{pmatrix}}_{\text{superclass correlation matrix}}; [\Phi]_{ij} = \langle \phi(\mathbf{x}_i), \phi(\mathbf{x}_j) \rangle = \begin{cases} 1, & i = j, \\ c, & y_i = y_j, i \neq j, \\ d, & y_i \neq y_j, h(y_i) = h(y_j), \\ 0, & h(y_i) \neq h(y_j), \end{cases}$$

Observation: Structure of Gram matrix in real dataset

- Diagonals: High correlation (c) within the same class
- Blocks: Low correlation (d) within the same superclass
- Others: Zero correlation between different superclasses



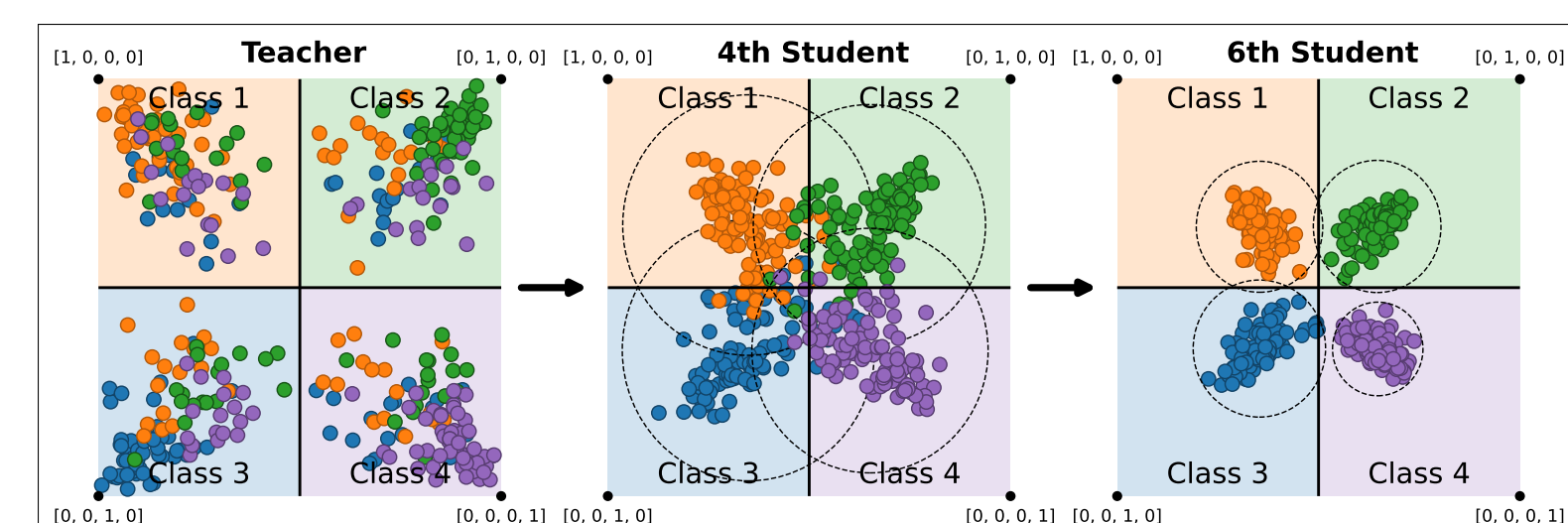
Top- K eigenvectors dominates in real world dataset.

SD Acts as a Classwise Output Clustering

From the Gram matrix assumption, $\mathbf{y}_i^{(t)}$ can be expressed as

$$\mathbf{y}_i^{(t)} = \underbrace{p^t \mathbf{e}(\hat{y}_i)}_{\text{given label}} + \underbrace{(q^t - p^t) \bar{\mathbf{e}}_{y_i}}_{\text{average label vectors over samples of the same class/superclass}} + \underbrace{(1 - r_s^t) \mathbf{1}_K / K}_{\text{uniform vector}},$$

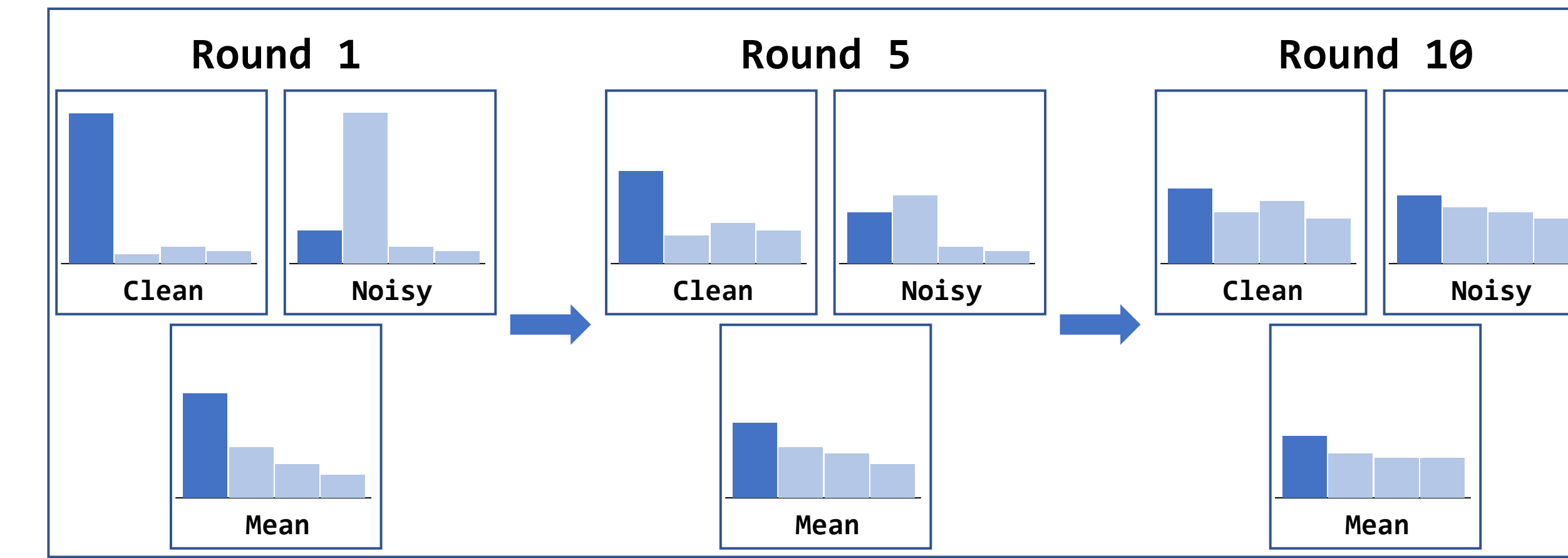
where $0 \leq p \leq q \leq r_s \leq 1$ is the function of c and d .



- The output of the distilled model is influenced by the mean output of the same class/superclass.
- As t increases, **the output aligns more with the mean output**.

The mean output of the same class/superclass induces label correction effects.

Gain of Multi-round SD in Label-Noise Scenarios



Confidence in clean samples decreases, while noisy samples gain confidence.

Theorem 4.1 (Gain of multi-round SD)

Suppose we have access to the population, i.e., $n \rightarrow \infty$. Let $\mathbf{C} \in \mathbb{R}^{K \times K}$ be the label corruption matrix of the training dataset, defined as $[\mathbf{C}]_{k,k'} := |\{i : y_i = k, \hat{y}_i = k'\}|/n$. Then, the t -th distilled model achieves 100% population accuracy if

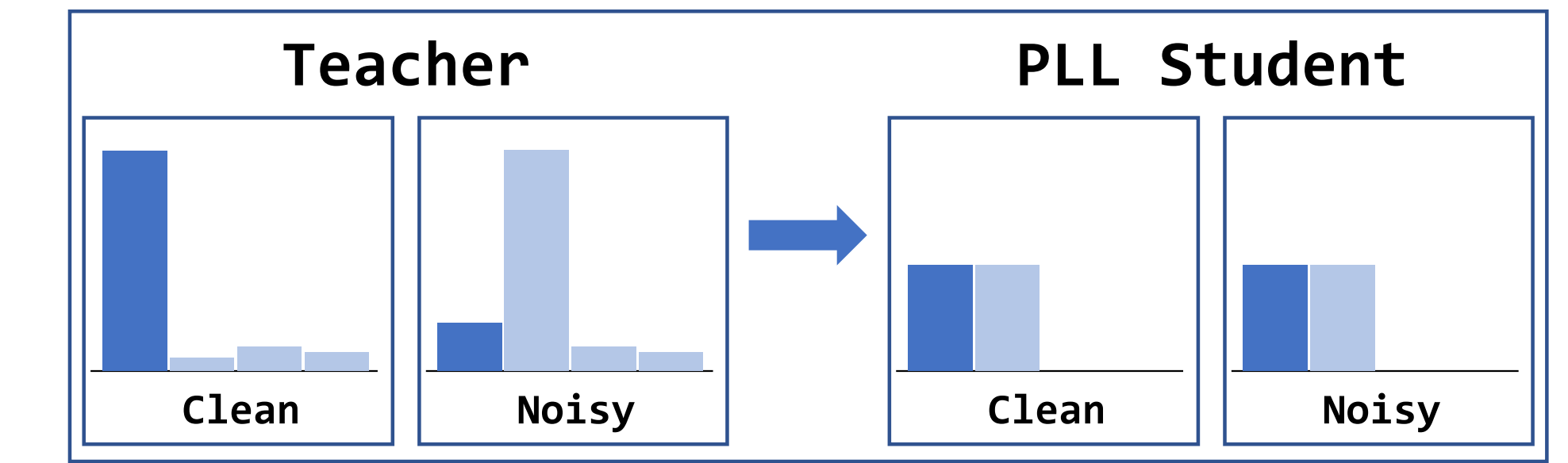
$$[\mathbf{C}]_{k,k} > [\mathbf{C}]_{k,k'} + 1 / \left((\tilde{\lambda}_K / \tilde{\lambda}_{K+1})^t - 1 \right), \quad \forall k, k' \in [K] \text{ with } k \neq k'$$

where $\tilde{\lambda}_K = \lambda_K / (K^2 n \lambda + \lambda_K)$ and $\tilde{\lambda}_{K+1} = \lambda_{K+1} / (K^2 n \lambda + \lambda_{K+1})$, with $\lambda_K = 1 - c + n(c - d)$ and $\lambda_{K+1} = 1 - c$.

- As t increases, the condition for achieving **100% accuracy becomes easier to satisfy**.
- **Advantages of multi-round SD:** When $[\mathbf{C}]_{k,k}$ and $[\mathbf{C}]_{k,k'}$ satisfy $[\mathbf{C}]_{k,k'} + 1 / \left((\tilde{\lambda}_K / \tilde{\lambda}_{K+1})^t - 1 \right) < [\mathbf{C}]_{k,k} < [\mathbf{C}]_{k,k'} + 1 / \left((\tilde{\lambda}_K / \tilde{\lambda}_{K+1})^{t-1} - 1 \right)$, t -th distilled model achieves 100% accuracy but $(t-1)$ -th model fails.
- **Multi-round SD enhances the model's robustness** through *label averaging effect*.

Soft Label Refinement using Top-Two Partial Labels

PLL student model: Targets two-hot label $\hat{y}_i$ by selecting top two labels from the teacher output



- Even with the output of the teacher model, the label averaging effect makes the output for the true label **at least the second-highest**, even for label-noisy samples.
- The top-two labels trade off clean sample confidence to **boost noisy sample confidence**.

Theorem 5.1 (Gain of PLL student model)

The PLL student model achieves 100% population accuracy if

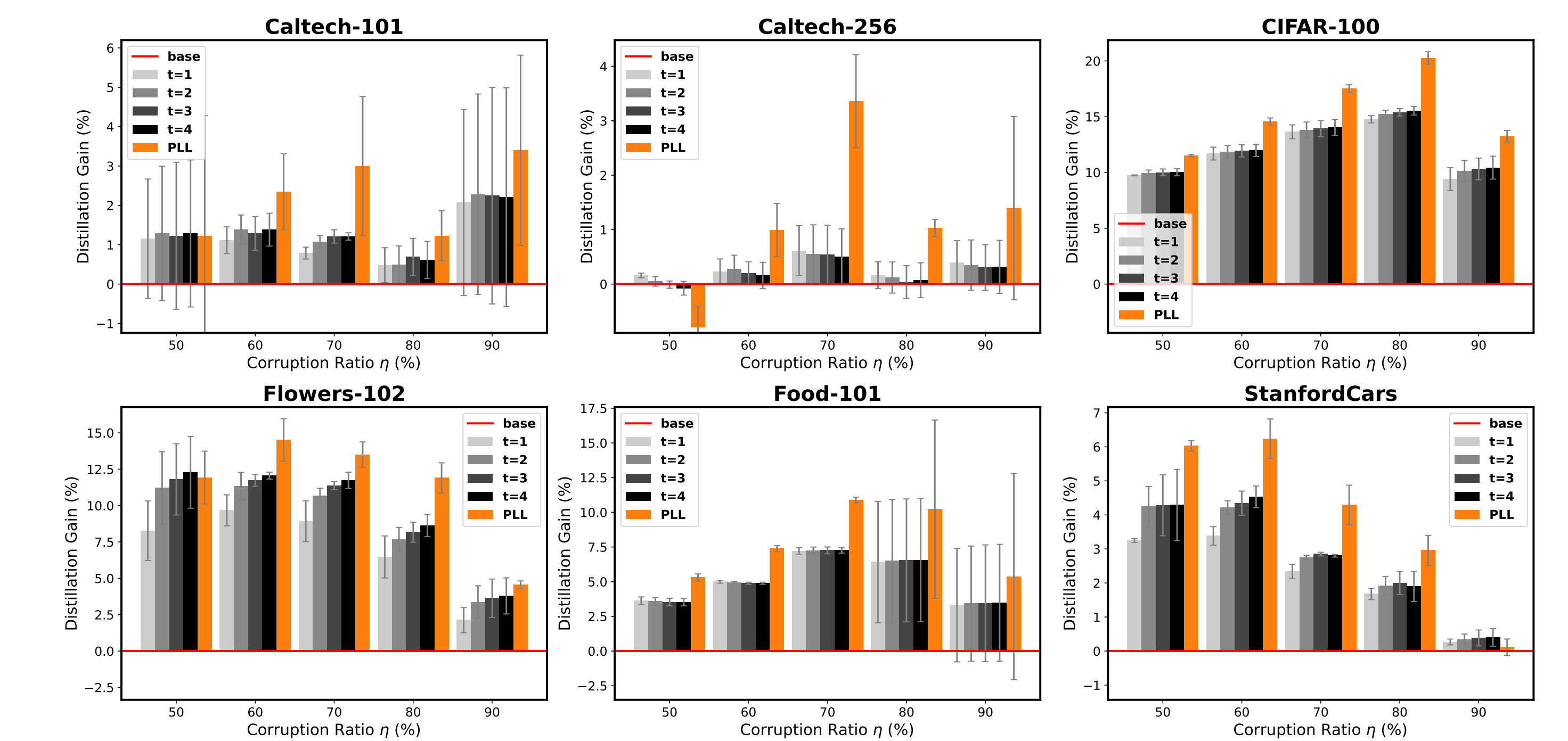
$$[\mathbf{C}]_{k,k} > [\mathbf{C}]_{k,k'}, \quad \forall k, k' \in [K] \text{ with } k \neq k',$$

where the label-corruption matrix $\mathbf{C}$ is defined by $[\mathbf{C}]_{k,k'} := |\{i : y_i = k, \hat{y}_i = k'\}|/n$.

PLL student model achieves 100% accuracy under milder conditions compared to multi-round SD.

Experiment on Real Datasets

- **Datasets:** CIFAR-100, Caltech-101/256, Flowers-102, Food-101, and StannfordCars.
- **Model:** ResNet34 networks pre-trained on ImageNet.
- **Label corruption:** Symmetric, Asymmetric, and Superclass corruption.



- Test accuracy **consistently improves** with each distillation round t .
- **PLL student model outperforms multi-round self-distillation models**.

References

- [1] Zeyuan Allen-Zhu and Yuanzhi Li. “Towards Understanding Ensemble, Knowledge Distillation and Self-Distillation in Deep Learning”. In: *The Eleventh International Conference on Learning Representations*. 2022.
- [2] Rudrajit Das and Sujay Sanghavi. “Understanding Self-Distillation in the Presence of Label Noise”. In: *Proceedings of the 40th International Conference on Machine Learning*. 2023.
- [3] Hossein Mobahi, Mehrdad Farajtabar, and Peter Bartlett. “Self-distillation amplifies regularization in hilbert space”. In: *Advances in Neural Information Processing Systems* 33 (2020), pp. 3351–3361.
- [4] Zhilu Zhang and Mert Sabuncu. “Self-distillation as instance-specific label smoothing”. In: *Advances in Neural Information Processing Systems* 33 (2020), pp. 2184–2195.